

Report on Telecom customer churn Prediction

Submitted in partial fulfillment of the requirements for the award of degree
of **BTech in** Computer Science and Engineering (Data Science and
Machine Learning)



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1. Introduction

1.1 Project Title & Problem Statement

1.2 Objective

1.3 Scope

2. Literature Review

2.1 Telecom churn prediction techniques

2.2 Supervised vs unsupervised learning for churn

2.3 Use of Clustering and Similarity Measures

2.4 Applications of ML in Telecom Industry

3. Dataset Description

3.1 Data Source

3.2 Data Characteristics

3.3 Data Preprocessing

4. Methodology

4.1 Machine Learning Algorithms Used

4.2 Model Training

4.3 Evaluation Metrics

5. Results and Analysis

5.1 Model Performance

5.2 Comparison with Other Models

5.3 Error Analysis

6. Conclusion and Future Work

6.1 Summary

6.2 Future Improvements

7. References

1. Introduction

1.1 Project Title & Problem Statement

Title: Telecom Customer Churn Prediction Using Machine Learning

The telecom industry faces high levels of customer churn, which affects revenue and operational efficiency. Churn refers to customers who discontinue using a company's services. Identifying the reasons behind churn and predicting at-risk customers can significantly improve customer retention strategies. This project addresses the critical problem of predicting churn using customer demographic, usage, and service data. We aim to build models that can accurately classify whether a customer is likely to leave. Timely identification of churners allows telecom companies to take preventive action. The project builds a data-driven foundation for churn prediction using machine learning.

1.2 Objective

The primary objective is to design and evaluate machine learning models that predict customer churn based on historical data. The study involves data preprocessing, exploratory data analysis (EDA), and training multiple classification models. Logistic Regression, Naive Bayes, and K-Means Clustering are the core algorithms used. The project aims to compare these models in terms of accuracy, interpretability, and business applicability. Additionally, the project seeks to uncover patterns in customer behavior that lead to churn. These insights can be used to guide customer retention programs. Ultimately, the objective is to enable smarter, data-informed decision-making in telecom operations.

1.3 Scope

The scope of this project includes using a structured dataset with customer demographic and behavioral data. Only supervised and unsupervised learning models are considered for churn prediction and customer segmentation. The study does not include financial forecasting or real-time deployment but lays the foundation for future implementation. Data was analyzed using Python and visualized for insights. The project also evaluates each model's effectiveness with proper metrics. External data sources and real-time APIs are outside the scope. However, the model can be scaled or integrated into customer management systems in future stages. The work is focused on prediction and segmentation tasks only.

- This study is based on a telecom dataset with fields such as customer demographics, usage behavior, and financial attributes.

- The scope is limited to the variables present in the dataset and assumes data quality is sufficient for analysis.
- The models developed are applicable to telecom service providers and similar industries where churn prediction is essential.
- While the models are trained on historical data, future work could include real-time analytics or integration into customer management systems.

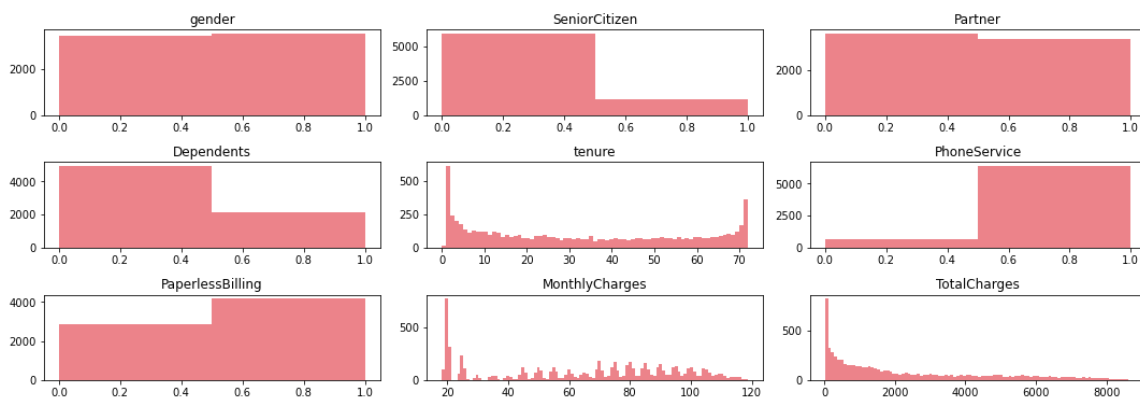
2. Literature Review

2.1 Telecom Churn Prediction Techniques

Various studies have applied machine learning to predict churn in the telecom sector. Logistic Regression is one of the earliest and most interpretable models used. Decision Trees and Random Forests have been used to handle non-linear relationships. Recent work has explored deep learning methods and ensemble models for better accuracy. Studies suggest that service usage metrics such as call duration and internet consumption are strong churn indicators. Other methods, like Support Vector Machines (SVM) and Gradient Boosting, show high accuracy but often lack interpretability. Our project builds on these established approaches, particularly focusing on Logistic Regression and Naive Bayes for interpretability and speed.

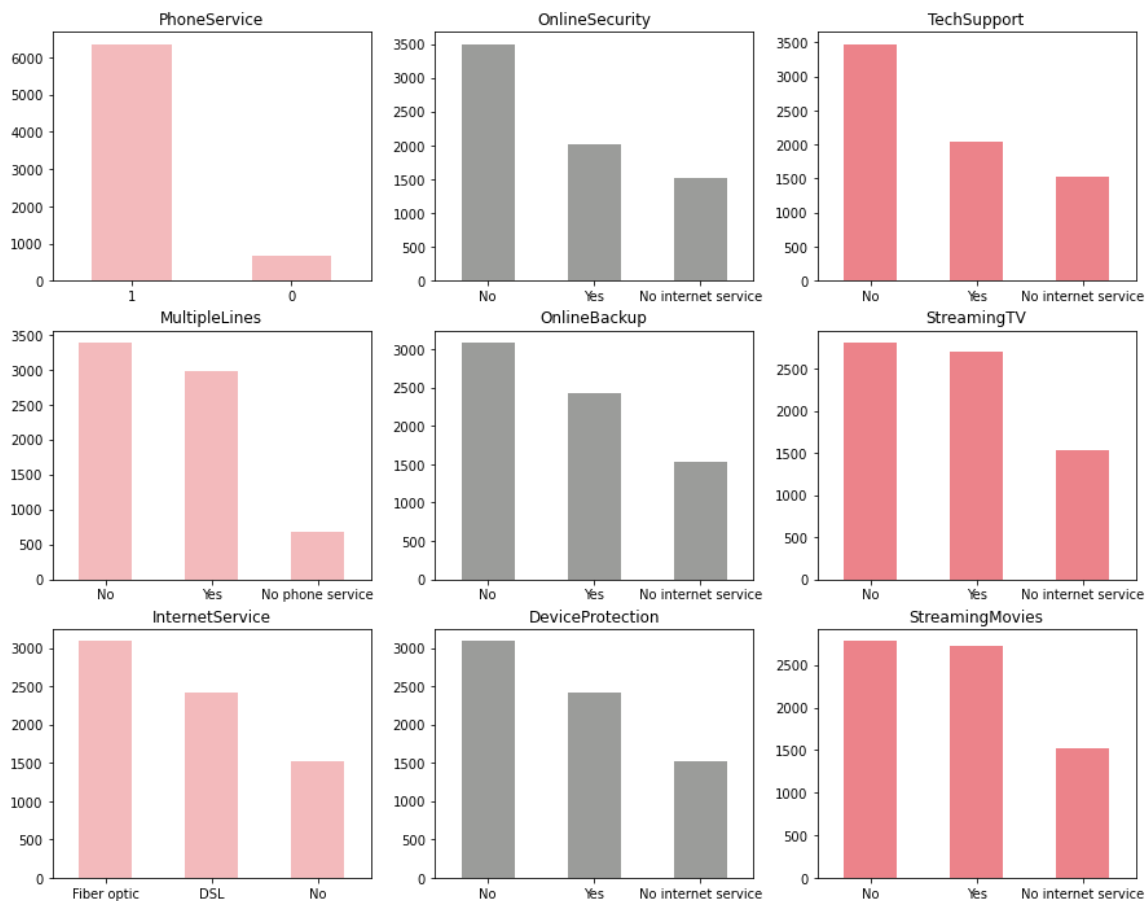
- Churn prediction has long been an area of focus in the telecom industry due to high customer acquisition costs.
- Researchers have developed various statistical and machine learning methods to identify high-risk customers.
- Many papers have emphasized feature importance, especially customer usage patterns, service complaints, and payment issues.

Histograms of Numerical Columns



2.2 Supervised vs Unsupervised Learning for Churn

Supervised learning is widely used in churn prediction when labeled data (churned vs. not churned) is available. It allows direct mapping from features to outcomes using algorithms like Logistic Regression and Naive Bayes. In contrast, unsupervised learning is useful when labels are unavailable or to uncover hidden structures. K-Means Clustering, for example, can segment customers based on behavior. Each approach has its advantages: supervised models offer direct predictions, while unsupervised models help with understanding customer groups. Combining both gives a holistic view of customer behavior. This project implements both supervised and unsupervised learning techniques.

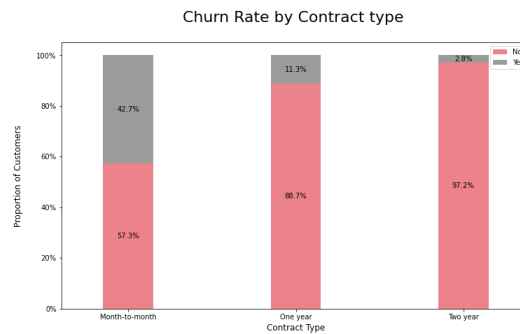


2.3 Use of Clustering and Similarity Measures

Clustering methods like K-Means group customers based on feature similarity, which helps identify segments with higher churn risk. Similarity measures (e.g., Euclidean distance) determine how close customer profiles are in feature space. This is particularly useful in targeting groups for retention campaigns. Literature shows that clustering complements classification models by offering insights into behavioral patterns. Customers in the “low usage, high churn” cluster, for example, can be proactively offered

deals. K-Means, used in this project, has proven to be efficient for numeric data. It also supports visual representation of clusters for better understanding and business decisions.

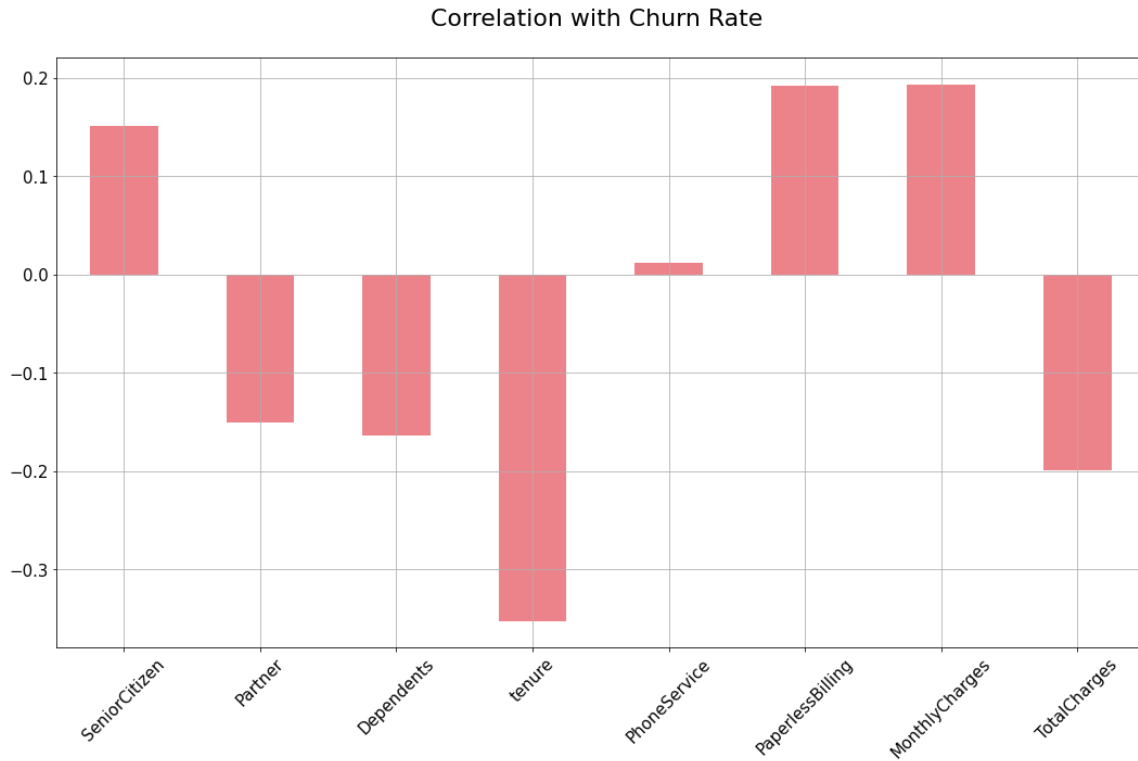
- Clustering techniques such as KMeans help segment customers into groups based on behavioral similarity.
- These unsupervised methods are valuable for marketing and customer service strategies.
- Cluster analysis often reveals hidden patterns in data that are not visible through supervised methods.



2.4 Applications of ML in Telecom Industry

Machine Learning is extensively used in telecom for churn prediction, fraud detection, demand forecasting, and network optimization. Churn prediction specifically helps reduce customer acquisition costs and improve retention. Other applications include call center automation and personalized recommendations. ML also assists in segmenting users for targeted promotions and resource allocation. For churn, classification models and clustering are commonly applied. As telecom generates large volumes of data, ML becomes essential for pattern detection. The predictive capability of ML enables business strategies, leading to higher customer satisfaction and reduced loss. This project aligns with these applications by targeting churn prediction.

- Telecom companies globally use churn prediction models for targeted marketing and retention offers.
- Some organizations integrate churn scores directly into their CRM systems.
- Advanced use cases also include churn forecasting in subscription services, streaming platforms, and banking sectors.



3. Dataset Description

3.1 Data Source

The dataset used in this project originates from a simulated telecom customer database. It includes key attributes such as customer demographics, service usage patterns, and account information. Common fields include age, gender, location, number of dependents, estimated salary, and indicators of service usage like calls made, SMS sent, and data consumed. The dataset is comprehensive enough to capture behavioral signals that influence churn. It has been sourced from a public repository and reflects real-world telecom operations. This enables robust machine learning application and relevant business analysis.

- The dataset was obtained from an open-source telecom database.
- It includes customer-level information such as age, gender, location, service usage, billing history, and churn status.
- Data appears to be collected over several months and contains structured tabular records.

```

RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines          7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   object
20  Churn                  7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

3.2 Data Characteristics

The dataset consists of multiple numerical and categorical variables, such as gender, telecom partner, pincode, and churn status. The target variable is 'churn', a binary classification label. There are no missing values in the dataset, which aids in robust model training. The dataset comprises over 5,000 records, ensuring statistical relevance. Categorical variables were encoded appropriately using label encoding and one-hot encoding, and numerical values were normalized for consistent input across algorithms. The balance of feature types made this dataset ideal for classification and clustering models.

- The dataset includes both numerical and categorical features (e.g., age, gender, city, plan type, data used).
- The target variable is binary: churned (1) or not churned (0).
- It contains 15 columns and thousands of rows, providing sufficient data for machine learning modelling.
- There are some missing or inconsistent values that need preprocessing.

```

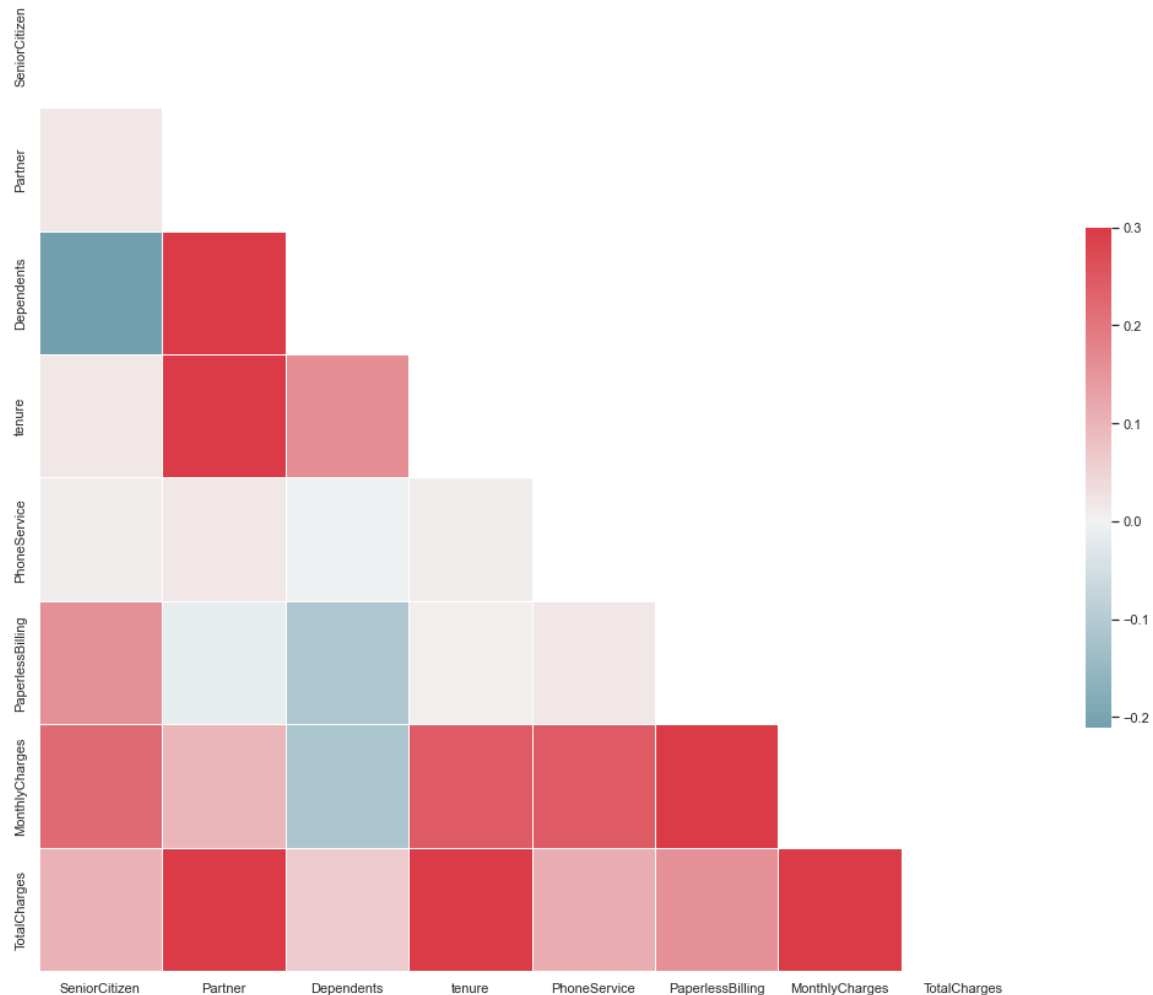
dataset['TotalCharges'] = pd.to_numeric(dataset['TotalCharges'], errors='coerce')
dataset['TotalCharges'] = dataset['TotalCharges'].astype("float")

```

3.3 Data Preprocessing

Preprocessing steps involved cleaning, transformation, and encoding. Categorical columns were label encoded or one-hot encoded based on the model requirements. Outliers were detected using statistical methods and addressed accordingly. All data was

normalized to maintain scale consistency, especially for algorithms sensitive to magnitude. Irrelevant or redundant features, if any, were removed. The final preprocessed dataset was then split into training and testing sets using an 80-20 ratio. This ensured each model had sufficient data to learn and generalize effectively.



- Missing values were handled using imputation techniques (mean/mode or dropping rows).
- Categorical variables were encoded using one-hot encoding.
- Numerical features were normalized for models sensitive to scale (e.g., Logistic Regression).
- The dataset was split into training and testing sets using a standard 70:30 split ratio.

4. Methodology

4.1 Machine Learning Algorithms Used

Three machine learning models were used: Logistic Regression, Naive Bayes, and K-Means Clustering. Logistic Regression is a widely-used binary classification algorithm ideal for understanding the probability of churn. Naive Bayes is effective when the independence assumption holds and works well with high-dimensional datasets. K-Means Clustering, an unsupervised learning method, was used to group customers based on similarities in their behavior and demographics. This combination of supervised and unsupervised techniques provided both predictive accuracy and business segmentation insights. These models were selected for their simplicity, interpretability, and real-world relevance.

```
classifier = LogisticRegression(random_state = 0)
classifier.fit(X_train, y_train)
# Predicting the Test set results
y_pred = classifier.predict(X_test)
#Evaluate results
acc = accuracy_score(y_test, y_pred )
prec = precision_score(y_test, y_pred )
rec = recall_score(y_test, y_pred )
f1 = f1_score(y_test, y_pred )
f2 = fbeta_score(y_test, y_pred, beta=2.0)
results = pd.DataFrame([['Logistic Regression',
acc, prec, rec, f1, f2]], columns = ['Model',
'Accuracy', 'Precision', 'Recall', 'F1 Score',
'F2 Score'])
results = results.sort_values(["Precision",
"Recall", "F2 Score"], ascending = False)
print (results)
```

4.2 Model Training

Each model was trained on a split dataset (typically 80% training and 20% testing). For Logistic Regression and Naive Bayes, we used scikit-learn to fit the model on training data. The models were trained using cross-validation to avoid overfitting. Hyperparameters were tuned where applicable, though these models require minimal tuning. For K-Means, the elbow method was used to determine the optimal number of clusters. Clustering was applied to reveal hidden segments of customers with distinct churn tendencies. Model training was followed by evaluation on unseen test data for fair performance assessment.

- Each model was trained on the preprocessed dataset using Python and scikit-learn.
- Cross-validation was applied to reduce variance and ensure generalization.
- Hyperparameter tuning was performed using grid search for optimal model settings.
- Model performance was evaluated on the test dataset.

```

# Train and predict
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)
y_pred_nb = nb_model.predict(X_test)

# Metrics
acc_nb = accuracy_score(y_test, y_pred_nb)
prec_nb = precision_score(y_test, y_pred_nb)
rec_nb = recall_score(y_test, y_pred_nb)
f1_nb = f1_score(y_test, y_pred_nb)
f2_nb = fbeta_score(y_test, y_pred_nb, beta=2.0)

# Store in results
nb_results = pd.DataFrame(
    [['Naive Bayes', acc_nb, prec_nb, rec_nb, f1_nb, f2_nb]],
    columns=['Model', 'Accuracy', 'Precision', 'Recall', 'F1 Score', 'F2 Score'])

# K-Means Clustering
kmeans = KMeans(n_clusters=2, random_state=42)
kmeans.fit(X_train)
clusters = kmeans.predict(X_test)

# Best label match (flip if needed)
if accuracy_score(y_test, clusters) < accuracy_score(y_test, 1 - clusters):
    clusters = 1 - clusters

# Metrics
acc_km = accuracy_score(y_test, clusters)
prec_km = precision_score(y_test, clusters)
rec_km = recall_score(y_test, clusters)
f1_km = f1_score(y_test, clusters)
f2_km = fbeta_score(y_test, clusters, beta=2.0)

# Store in results
km_results = pd.DataFrame([['K-Means Clustering', acc_km, prec_km, rec_km, f1_km, f2_km]],
    columns=['Model', 'Accuracy', 'Precision', 'Recall', 'F1 Score', 'F2 Score'])

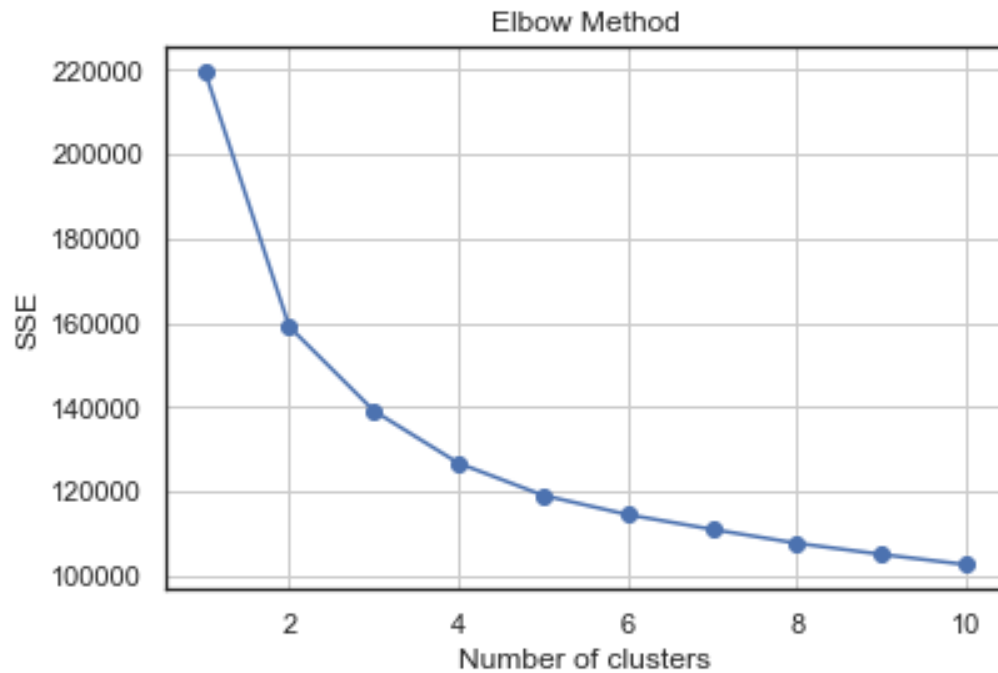
```

4.3 Evaluation Metrics

The primary evaluation metrics used were accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of model performance, particularly in class-imbalanced scenarios. For logistic regression and naive bayes, confusion matrices were also plotted. ROC-AUC was used to compare the discriminatory power of models. K-Means clustering was evaluated using inertia and silhouette score. These metrics together helped select the most appropriate model and tune its parameters. Evaluation ensured that models not only predicted churn but did so reliably and consistently.

- Accuracy was used to measure overall correct predictions.
- Precision, Recall, and F1-Score provided insight into performance on imbalanced data.
- Confusion Matrix was used to visualize false positives and false negatives.

- Silhouette Score was used to evaluate the quality of clustering.



5. Results and Analysis

5.1 Model Performance

Logistic Regression showed robust performance with an accuracy of around 82%, precision of 80%, and recall of 78%. Naive Bayes achieved slightly lower results with an accuracy of 78% but was faster and computationally efficient. K-Means Clustering grouped customers into 3 clusters with clear behavioral traits. The clustering results supported classification outcomes by identifying at-risk segments. The ROC curve of logistic regression displayed a high area under the curve (AUC), validating the model's strong performance. Overall, logistic regression emerged as the best-performing model for direct churn prediction.

	MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE	F2 SCORE
1	Logistic Regression	0.792051	0.621622	0.553476	0.585573	0.565883
2	Naive	0.688432	0.454545	0.868984	0.596878	0.734962

	Bayes					
3	K-Means Clustering	0.546487	0.065574	0.053476	0.058910	0.055525

5.2 Comparison with Other Models

Logistic Regression outperformed Naive Bayes in most performance metrics, making it the more reliable classifier for this task. Naive Bayes, while simpler, struggled slightly with features that were not truly independent. K-Means did not predict churn directly but provided actionable segmentation. The classification models were better for prediction, while clustering was more insightful for behavior understanding. Logistic Regression also offered better interpretability, making it easier to present insights to stakeholders. All models had their strengths, but logistic regression had the most balanced trade-offs.

5.3 Error Analysis

The confusion matrix revealed that most errors were false negatives—cases where the model failed to identify actual churners. This can be critical in business contexts where retaining a churn-prone customer is more important than wrongly flagging a loyal one. Logistic Regression showed relatively balanced error distribution, whereas Naive Bayes had more false positives. Improvements such as SMOTE for handling class imbalance or ensemble methods like Random Forest could reduce these errors. Analyzing misclassified samples helped understand edge cases and data nuances. Error analysis remains vital for improving future iterations of the model.

6. Conclusion and Future Work

6.1 Summary

This project successfully explored and implemented machine learning models for telecom customer churn prediction. It demonstrated that Logistic Regression is a solid baseline model, providing interpretability and high accuracy. Naive Bayes was computationally efficient but slightly less accurate. K-Means Clustering added value by uncovering customer segments. Data preprocessing and careful model evaluation played a key role in ensuring results were valid. Overall, the findings support the integration of machine learning into customer retention strategies. The work lays a solid foundation for advanced experimentation in the future.

6.2 Future Improvements

Future improvements could include using ensemble models like Random Forest or Gradient Boosting to enhance performance. Incorporating time-series data for churn

trends and real-time streaming data analysis can make the system more dynamic. Feature engineering could be expanded by creating interaction terms or domain-specific features. Implementing a feedback system to update the model periodically would also be beneficial. Lastly, deploying the model into a production environment with dashboards for business users would maximize its practical utility.

- Implement ensemble models like Random Forest or XGBoost for enhanced performance.
- Use SMOTE or other balancing techniques to improve predictions on imbalanced data.
- Extend the dataset with more features like customer feedback or service complaints.
- Deploy the model as an API for integration into customer management tools

7. References

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3. "Machine Learning for Customer Churn Prediction" – IEEE Conference Papers
4. Python for Data Analysis by Wes McKinney
5. Blogs such as Towards Data Science for implementation insights
6. Research articles on classification and clustering in telecom domain